

# MediLink Healthcare Appointment Platform

## Agile Delivery Plan (v2.0) — 8-Week MVP + Phase 2 Roadmap

|             |                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Document    | Agile Delivery Plan v2.0 (8-week MVP)                                                                                                  |
| Date        | 2026-06-16                                                                                                                             |
| Prepared By | Satyam Kumar                                                                                                                           |
| Cadence     | 8-week sprint cycle • weekly demos • 4 two-week sprints                                                                                |
| Scope       | Patient Mobile App (Flutter) + Patient Web App (Next.js 15). Healthcare Facility App, Doctor Portal and Admin Portal are OUT of scope. |
| Stack       | Flutter (mobile) • Next.js 15 web + Route Handlers (backend) • Supabase PostgreSQL / Auth / Storage / Realtime • Vercel • EN/AR (RTL)  |
| Status      | For stakeholder review & approval                                                                                                      |

**Alignment note:** This revision aligns to the client requirement — “*Development (Agile Sprints): 8-week sprint cycle with weekly demos.*” The 8 weeks are delivered as four 2-week sprints with a demo at the end of every week (8 demos). Everything outside the agreed MVP list is moved to Phase 2 (Post-MVP Enhancements).

### 1. Executive Summary

MediLink delivers two patient-facing products — a Flutter mobile app and a Next.js 15 web app — on a single Supabase backend (PostgreSQL + Auth + Storage + Realtime), deployed on Vercel + Supabase. Phase 1 ships a complete, revenue-generating patient booking platform in **8 weeks**; Phase 2 layers on engagement, telemedicine and AI features after launch.

**Phase 1 (MVP — 8 weeks):** the full patient journey — register → find a doctor → book → pay (Thawani) → get invoice & notifications — on web and mobile, bilingual (EN/AR RTL).

**Cadence:** four 2-week sprints; a working demo every Friday (8 weekly demos) keeps stakeholders aligned.

**Revenue first:** payments and invoices land inside the MVP; AI and telemedicine are explicitly Phase 2.

**Critical dependency:** patients book against doctor/facility availability. Since provider/admin portals are out of scope, that data is seeded and admin-managed externally (addressed in Sprint 1 setup).

### 2. MVP Definition (Phase 1)

The MVP is the minimum that lets a patient find a doctor, book, pay, and get confirmation on both platforms. Scope is fixed to the agreed list below — nothing more.

| MVP Capability      | What it includes                                                           |
|---------------------|----------------------------------------------------------------------------|
| Authentication      | Email registration, login, OTP verification, Google login, forgot password |
| Patient Profile     | Create / edit profile, basic medical history                               |
| Doctor Discovery    | Search + filter doctors, doctor profile, facility profile                  |
| Appointment Booking | Book appointment with overbooking / race-condition control + buffer time   |
| Reschedule / Cancel | Reschedule and cancel with atomic slot release                             |
| Web Check-In        | View appointment status / check-in                                         |
| Thawani Payment     | Thawani checkout + webhook reconciliation                                  |

| MVP Capability       | What it includes                                      |
|----------------------|-------------------------------------------------------|
| Invoices             | Invoice generation + payment history                  |
| Notifications        | Email + push + in-app (confirmation / reminder)       |
| English + Arabic RTL | Full bilingual UI with RTL layout on every MVP screen |

**MVP success metrics:** ≥1 completed paid booking per platform; booking funnel < 90s; payment success ≥ 95%; zero double-bookings under concurrency test; EN/AR verified on all screens.

### 3. Roadmap — Two Phases

| PHASE 1 — MVP (8 Weeks)                                                                                                                 | PHASE 2 — Post-MVP Enhancements                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Auth • Patient Profile • Doctor Discovery • Booking • Reschedule/Cancel • Web Check-In • Thawani • Invoices • Notifications • EN/AR RTL | Family Members • Document Vault • Insurance • Ratings & Reviews • Follow-up Booking • Multi-location Booking • Ompay • Apple Login • SMS • Telemedicine • AI Symptom Checker • AI Doctor Suggestions • AI Scheduling Assistant • Analytics • Audit Logs • Offline Mode • Dark Mode |

### 4. Timeline Table (8 Weeks, Weekly Demos)

| Week | Sprint   | Focus                                            | Friday Demo Milestone                                   |
|------|----------|--------------------------------------------------|---------------------------------------------------------|
| 1    | Sprint 1 | Setup, schema, CI/CD, seed data; auth API start  | Skeleton apps deploy to staging; seeded doctors visible |
| 2    | Sprint 1 | Auth UI (email, OTP, Google, reset) web + mobile | Live login + OTP on both apps                           |
| 3    | Sprint 2 | Patient profile + medical history                | Edit profile end-to-end (EN/AR)                         |
| 4    | Sprint 2 | Doctor discovery: search, filter, profiles       | Search + view doctor/facility profile                   |
| 5    | Sprint 3 | Booking with slot locking                        | Book an appointment (unpaid)                            |
| 6    | Sprint 3 | Reschedule, cancel, web check-in                 | Full booking lifecycle + check-in                       |
| 7    | Sprint 4 | Thawani payment + invoices                       | Paid booking + invoice download                         |
| 8    | Sprint 4 | Notifications, EN/AR hardening, UAT, launch      | MVP launch candidate; UAT sign-off                      |

**MVP go-live:** end of Week 8. Each weekly demo is a working, deployable increment — not slides.

### 5. Team Structure (small team, ~6–7 people)

| Role                             | Count | Responsibility                                                           |
|----------------------------------|-------|--------------------------------------------------------------------------|
| Tech Lead / Architect            | 1     | Supabase schema + RLS, Route Handler (backend) design, reviews, releases |
| Next.js Engineer (Web + Backend) | 2     | Web app + Route Handlers (shared API for both clients)                   |
| Flutter Engineer                 | 2     | Mobile app, shared API consumption                                       |
| QA Engineer                      | 1     | Test plans, concurrency tests, weekly-demo verification, UAT             |
| PM / Scrum Master                | 1     | Backlog, weekly demos, stakeholder comms, UAT sign-off                   |
| UI/UX Designer (0.5)             | 0.5   | Designs one sprint ahead; shadcn / Figma kit                             |

**Velocity:** ~40 story points / 2-week sprint → ~160 points across the 8-week MVP.

## 6. 8-Week Sprint Plan

Four 2-week sprints. Each sprint: Objective • User Stories (points) • Acceptance Criteria • Risks • Deliverables. Platform tags: [W] web, [M] mobile, [S] shared API/DB. Demo every Friday.

### Sprint 1 — Foundation & Authentication

**Weeks:** Weeks 1-2 | **Story Points:** 40

**Weekly demos:** W1: skeleton + seeded data demo. W2: live auth on both apps.

**Objective:** Stand up both apps, the Supabase schema with RLS, CI/CD and seeded provider data, then deliver full authentication (email, OTP, Google, reset) on web and mobile.

#### User Stories

| ID         | Story                                                                                                                                  | Pts |
|------------|----------------------------------------------------------------------------------------------------------------------------------------|-----|
| S1-1 [S]   | Supabase schema + RLS (profiles, patient_profiles, doctors, facilities, availability, appointments, payments, invoices, notifications) | 8   |
| S1-2 [S]   | Seed scripts / admin-managed doctors, facilities, availability (stand-in for out-of-scope portals)                                     | 5   |
| S1-3 [W]   | Next.js 15 scaffold (App Router, Tailwind, shadcn/ui, next-intl EN/AR) + CI to Vercel                                                  | 4   |
| S1-4 [M]   | Flutter scaffold (routing, state, supabase_flutter, EN/AR ARB) + CI build                                                              | 4   |
| S1-5 [S]   | Auth API: email signup/login + session (Supabase Auth)                                                                                 | 4   |
| S1-6 [S]   | OTP verification flow                                                                                                                  | 4   |
| S1-7 [W/M] | Google OAuth + forgot/reset password                                                                                                   | 5   |
| S1-8 [W/M] | Auth UI (register/login/OTP/reset) both platforms + RTL                                                                                | 6   |

#### Acceptance Criteria

- Both apps build and deploy to staging via CI; RLS on every table (patient reads only own rows).
- Seed produces  $\geq 10$  doctors,  $\geq 3$  facilities and bookable slots.
- New user can register, verify OTP, and log in (email + Google) on both apps; reset works.
- Arabic flips layout to RTL on all auth screens.

#### Risks

- No provider portal  $\rightarrow$  availability data gap. **Mitigation:** seed + admin scripts; lock data contract in W1.
- OAuth deep-link / redirect config differs web vs mobile. **Mitigation:** configure + device-test early.

#### Deliverables

Deployed skeleton (web + mobile), schema + RLS + seed, CI/CD, and working authentication on both platforms.

### Sprint 2 — Patient Profile & Doctor Discovery

**Weeks:** Weeks 3-4 | **Story Points:** 40

**Weekly demos:** W3: profile management demo. W4: doctor search + profiles demo.

**Objective:** Patient can manage their profile and basic medical history, then search/filter doctors and view doctor & facility profiles (SEO-rendered on web).

#### User Stories

| ID         | Story                                                            | Pts |
|------------|------------------------------------------------------------------|-----|
| S2-1 [S]   | Patient profile API (CRUD) + basic medical history               | 5   |
| S2-2 [W/M] | Profile management screens both platforms (EN/AR)                | 6   |
| S2-3 [S]   | Doctor search + filter API (specialty, location, availability)   | 5   |
| S2-4 [W]   | Web search/filter UI + SEO-rendered doctor pages (SSR + JSON-LD) | 8   |
| S2-5 [M]   | Mobile search/filter UI                                          | 5   |
| S2-6 [W/M] | Doctor profile + facility profile screens                        | 6   |
| S2-7 [S]   | i18n strings wired for profile + discovery                       | 2   |
| S2-8 [QA]  | Test pass: profile + discovery (EN/AR, cross-device)             | 3   |

## Acceptance Criteria

- bullet Patient can create/edit profile + medical history; persists across sessions.
- bullet Search returns filtered doctors; web doctor pages server-rendered with valid structured data.
- bullet Doctor and facility profiles display correctly on web and mobile, both languages.

## Risks

- bullet SEO rendering regressions. **Mitigation:** SSR snapshot tests in CI.
- bullet Filter performance on larger seed data. **Mitigation:** indexed queries, pagination.

## Deliverables

Profile management + full doctor/facility discovery on both platforms, SEO-rendered web pages.

# Sprint 3 — Appointment Booking, Reschedule/Cancel & Web Check-In

**Weeks:** Weeks 5-6 | **Story Points:** 41

**Weekly demos:** W5: book-an-appointment demo. W6: full lifecycle + check-in demo.

**Objective:** Deliver the core booking lifecycle with overbooking protection: book, reschedule, cancel, and web check-in (payment added in Sprint 4).

## User Stories

| ID         | Story                                                                           | Pts |
|------------|---------------------------------------------------------------------------------|-----|
| S3-1 [S]   | Booking API with transactional slot locking (overbooking control + buffer time) | 8   |
| S3-2 [S]   | Reschedule API (atomic old-slot release)                                        | 4   |
| S3-3 [S]   | Cancel API + slot release                                                       | 3   |
| S3-4 [W/M] | Booking flow UI (select slot -> confirm) both platforms                         | 8   |
| S3-5 [W/M] | Reschedule + cancel UI                                                          | 5   |
| S3-6 [W/M] | Web check-in (view status / check-in)                                           | 5   |
| S3-7 [QA]  | Concurrency / double-booking test suite                                         | 5   |
| S3-8 [S]   | Booking confirmation events (hook for notifications)                            | 3   |

## Acceptance Criteria

- bullet Two concurrent bookings on the same slot → exactly one succeeds (no double-book).

- bullet Reschedule frees the old slot atomically; cancel releases the slot.
- bullet Booking completes in < 90s in usability test; web check-in shows correct status.
- bullet Buffer time enforced between appointments.

## Risks

- bullet Race conditions under load — highest technical risk. **Mitigation:** DB unique constraint + FOR UPDATE + concurrency suite.
- bullet Edge cases in reschedule/cancel state. **Mitigation:** state-machine tests.

## Deliverables

Full booking lifecycle (book / reschedule / cancel) + web check-in on both platforms, with concurrency test report.

# Sprint 4 — Payments, Invoices, Notifications & Launch

**Weeks:** Weeks 7-8 | **Story Points:** 39

**Weekly demos:** W7: paid booking + invoice demo. W8: MVP launch candidate + UAT sign-off.

**Objective:** Close the revenue loop with Thawani, generate invoices, deliver multi-channel notifications, harden EN/AR, run UAT and ship the MVP.

## User Stories

| ID         | Story                                                                  | Pts |
|------------|------------------------------------------------------------------------|-----|
| S4-1 [S]   | Thawani checkout + webhook (payment status -> appointment), idempotent | 8   |
| S4-2 [S]   | Invoice generation (PDF + record) + payment history API                | 5   |
| S4-3 [W/M] | Payment UI + confirmation + payment history screen                     | 6   |
| S4-4 [S]   | Notification service: email (Resend) + push (FCM/APNs) + in-app store  | 7   |
| S4-5 [W/M] | In-app notification center + push handling                             | 4   |
| S4-6 [S]   | Slot-hold + payment-timeout release worker                             | 3   |
| S4-7 [QA]  | MVP regression, EN/AR full sweep, cross-device, UAT                    | 4   |
| S4-8 [S]   | Production launch checklist + rollout                                  | 2   |

## Acceptance Criteria

- bullet Booking -> Thawani payment -> confirmed; webhook reconciles; failed/expired payment releases the held slot.
- bullet Invoice generated, downloadable, and listed in payment history.
- bullet Confirmation via email + push + in-app; reminder scheduled.
- bullet Payment success ≥ 95% in staging; idempotent webhook (no double-charge/confirm).
- bullet All MVP acceptance criteria pass; UAT signed off.

## Risks

- bullet Payment webhook reliability / idempotency. **Mitigation:** idempotency keys, signature verification, reconciliation job.
- bullet Launch-week scope creep. **Mitigation:** feature freeze after W7; W8 is hardening + UAT only.

## **Deliverables**

PRODUCTION MVP - paid booking journey live on web + mobile, invoices, payment history, notifications, bilingual. Go-live sign-off.

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## 7. Phase 2 Roadmap (Post-MVP Enhancements)

Sequenced after MVP launch, grouped into themed releases. Indicative; re-prioritised with stakeholders post-launch using real usage data.

| Release | Theme                | Features                                                           | Indicative  |
|---------|----------------------|--------------------------------------------------------------------|-------------|
| 2.1     | Engagement & Records | Family Members, Document Vault, Insurance, Ratings & Reviews       | Weeks 9-11  |
| 2.2     | Booking & Payments+  | Follow-up Booking, Multi-location Booking, Ompay, Apple Login, SMS | Weeks 12-14 |
| 2.3     | Telemedicine         | Video Consultation, Virtual Waiting Room, Pre-consultation Form    | Weeks 15-17 |
| 2.4     | AI & Intelligence    | AI Symptom Checker, AI Doctor Suggestions, AI Scheduling Assistant | Weeks 18-20 |
| 2.5     | Platform & UX        | Analytics, Audit Logs, Offline Mode, Dark Mode                     | Weeks 21-22 |

**Note:** AI and telemedicine deliberately follow the revenue-generating core, per the prioritisation rule. AI features will ship with medical-safety guardrails and disclaimers (no definitive diagnosis).

## 8. Risks & Dependencies

| #  | Risk / Dependency                                 | Impact | Likely | Mitigation                                                 | Owner        |
|----|---------------------------------------------------|--------|--------|------------------------------------------------------------|--------------|
| D1 | Provider/availability data (portals out of scope) | High   | High   | Seed + admin scripts; data contract in Sprint 1            | Tech Lead    |
| D2 | Single shared API for 2 clients                   | High   | Med    | Lock Route Handler contract before feature work            | Tech Lead    |
| R1 | Booking race conditions / double-booking          | High   | Med    | DB unique index + FOR UPDATE + concurrency suite           | Web Eng/QA   |
| R2 | Thawani webhook idempotency / reconciliation      | High   | Med    | Idempotency keys, signature verify, reconciliation job     | Web Eng      |
| R3 | OAuth deep links (Google) web vs mobile           | Med    | Med    | Configure URI schemes early; device tests                  | Flutter Eng  |
| R4 | 8-week scope pressure                             | High   | Med    | Fixed MVP list; strict Phase-2 deferral; W8 hardening only | PM           |
| R5 | PHI exposure via Storage/data                     | High   | Low    | RLS on every table, signed URLs, private buckets           | Tech Lead    |
| R6 | Weekly-demo overhead vs delivery                  | Low    | Med    | Demo the working increment; no separate demo build         | Scrum Master |

## 9. Stakeholder Approval

This plan is submitted for review and sign-off. Approval confirms the fixed 8-week MVP scope above and the deferral of all Phase 2 features.

| Role                         | Name | Signature | Date |
|------------------------------|------|-----------|------|
| Client / Product Owner       |      |           |      |
| Delivery Lead (Satyam Kumar) |      |           |      |